Local Projections in Macroeconomics

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Basic LP

$$y_{t+h} = \alpha_h + \beta_h s_t + \gamma_h x_t + v_{t+h} \qquad h = 0, 1, \ldots, H$$

Notation:

- y_t: outcome variable
- s_t: treatment/intervention
- x_t: controls (including also lags of y_t and s_t)

What are we interested in? **Impulse response functions...**
$\rightarrow$ if s_t can be interpreted as a macro shock then
$\mathcal{R}_{s \rightarrow y}(h) = \beta_h$ i.e. $\{\beta_h\}_{h=0}^{H}$ are the IRFs

Romer and Romer (2004) I

- **Issue:** Conventional measures (e.g., federal funds rate) are endogenous and reflect the Fed's forward-looking behavior.
- Policy reacts systematically to expected future output and inflation $\Rightarrow$ biased estimates of monetary policy effects.

- **New measure of monetary policy shocks (1969–1996):**
  - Controls for the Fed's internal forecasts.
  - Removes systematic responses to anticipated economic developments.
  - Focuses on intended changes in the federal funds rate around FOMC meetings.

Romer and Romer (2004) II

- **Results:**
  - Monetary policy has large, rapid, and statistically significant effects on output and inflation.
  - Effects are stronger and faster than with previous measures.
- **Conclusion:** Earlier measures were likely contaminated by endogeneity and forward-looking behavior.

Romer and Romer (2004) III

Regression:

$$\Delta \mathrm{ff}_t = \alpha + \beta\, \mathrm{ffb}_t + \sum_{i=-1}^{2} \gamma_i\, \mathbf{E}_t[x_{t+i}] + \sum_{i=-1}^{2} \lambda_i \left(\mathbf{E}_t[x_{t+i}] - \mathbf{E}_{t-1}[x_{t+i}]\right) + \phi\, \mathbf{E}_t[u_t] + \epsilon_t$$

where

- $\Delta \mathrm{ff}_t$ is the change in the intended funds rate around FOMC meeting t.
- ffb_t is the level of intended funds rate before any changes associated with meeting t.
- $\mathbf{E}_t[\cdot]$ is the Fed forecast formulated at time t.
- x_t contains real GDP growth and inflation; and u_t denotes unemployment.